

Multiplicação por escalar, soma e subtração

01. Multiplicar a matriz $m = \begin{bmatrix} 9 & -7 \\ -3 & 5 \end{bmatrix}$ por -1 .

$$\begin{bmatrix} -1 * 9 & -1 * -7 \\ -1 * -3 & -1 * 5 \end{bmatrix} = \begin{bmatrix} -9 & 7 \\ 3 & -5 \end{bmatrix}$$

02. Subtrair a matriz $A = \begin{bmatrix} 9 & -7 \\ -3 & 5 \end{bmatrix}$ a $B = \begin{bmatrix} -7 & 3 \\ -5 & 8 \end{bmatrix}$

$$A - B = \begin{bmatrix} 9 - (-7) & -7 - 3 \\ -3 - (-5) & 5 - 8 \end{bmatrix} = \begin{bmatrix} 16 & -10 \\ 2 & -3 \end{bmatrix}$$

Matriz Transposta

01. $A = \begin{bmatrix} 3 & 5 \\ 6 & 7 \end{bmatrix};$

$$A^T = \begin{bmatrix} 3 & 6 \\ 5 & 7 \end{bmatrix}$$

02. $V = \begin{bmatrix} 7 \\ -2 \end{bmatrix};$

$$V^T = \begin{bmatrix} 7 & -2 \end{bmatrix}$$

$$03. v = [3 \ -1 \ 1];$$

$$v^T = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

04. O produto interno entre

$$a = [3 \ -2 \ 1] \quad \text{e} \quad b = [5 \ 2 \ 6]$$

$$a * b = (3 * 5) + (-2 * 2) + (1 * 6)$$

$$a * b = 15 - 4 + 6$$

$$\boxed{a * b = 17}$$

Multiplicações entre matrizes

$$01. A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{e} \quad B = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix};$$

$$AB = \begin{bmatrix} 1 * 0 + 2 * 2 & 1 * 1 + 2 * 0 \\ 3 * 0 + 4 * 2 & 3 * 1 + 4 * 0 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 8 & 3 \end{bmatrix}.$$

$$02. A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix};$$

$$AB = \begin{bmatrix} 0 \times 1 + 1 \times 3 & 0 \times 2 + 1 \times 4 \\ 2 \times 1 + 0 \times 3 & 2 \times 2 + 0 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 2 & 4 \end{bmatrix}.$$

$$03. A = \begin{bmatrix} 4 & 2 & 5 \\ 3 & 4 & 6 \\ 4 & -1 & -7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 6 \\ 2 & 0 & -1 \\ 3 & -1 & 10 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 \times 1 + 2 \times 2 + 5 \times 3 & 4 \times 1 + 2 \times 0 + 5 \times (-1) & 4 \times 6 + 2 \times (-1) + 5 \times 10 \\ 3 \times 1 + 4 \times 2 + 6 \times 3 & 3 \times 1 + 4 \times 0 + 6 \times (-1) & 3 \times 6 + 4 \times (-1) + 6 \times 10 \\ 4 \times 1 + (-1) \times 2 + (-7) \times 3 & 4 \times 1 + (-1) \times 0 + (-7) \times (-1) & 4 \times 6 + (-1) \times (-1) + (-7) \times 10 \end{bmatrix}$$

$$AB = \begin{bmatrix} 23 & -1 & 72 \\ 29 & -3 & 74 \\ -19 & 11 & -45 \end{bmatrix}$$

Matriz identidade e inversa

2x2

$$01. A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

Diagram showing the calculation of the determinant: $1 \times 0 - 2 \times 0 = 0$. Arrows point from the elements 1, 2, 0, and 0 to the result 0.

$\det(A) = 0$ (não possui inversa)

$$02. A = \begin{bmatrix} 1 & 5 \\ 2 & 7 \end{bmatrix}$$

Diagram showing the calculation of the determinant: $1 \times 7 - 5 \times 2 = -3$. Arrows point from the elements 1, 5, 2, and 7 to the result -3.

$$\det(A) = 7 - 10 = -3$$

$$A^{-1} = \frac{1}{-3} \begin{bmatrix} 7 & -5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{7}{3} & \frac{5}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

$$03. A = \begin{bmatrix} 6 & 4 \\ 3 & 2 \end{bmatrix}$$

Diagram showing the calculation of the determinant: $6 \times 2 - 4 \times 3 = 0$. Arrows point from the elements 6, 4, 3, and 2 to the result 0.

$\det(A) = 0$ (não possui inversa)

$$04. A = \begin{bmatrix} 3 & 5 \\ 2 & -7 \end{bmatrix}$$

Diagram showing the calculation of the determinant: $3 \times (-7) - 5 \times 2 = -21 - 10 = -31$. Arrows point from the elements 3, 5, 2, and -7 to the result -31.

$$\det(A) = -21 - 10 = -31$$

$$A^{-1} = \frac{1}{-31} \begin{bmatrix} -7 & -5 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} \frac{7}{31} & \frac{5}{31} \\ \frac{2}{31} & -\frac{3}{31} \end{bmatrix}$$

3x3

01. $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 4 & 2 & 5 \end{bmatrix}$

2 1
1 0
4 2

0 4 5 0 4 6

$$\det(A) = 0 + 4 + 6 - 5 - 4 - 0$$

$$\det(A) = 1$$

$$C_{21} = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} = -(-1) = 1$$

6 5

$$C_{22} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = -2$$

12 10

$$C_{23} = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = -(0) = 0$$

4 4

$$C_{11} = \begin{bmatrix} 0 & 1 \\ 2 & 5 \end{bmatrix} = -2$$

2 0

$$C_{12} = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} = -(1) = -1$$

4 5

$$C_{13} = \begin{bmatrix} 1 & 0 \\ 4 & 2 \end{bmatrix} = 2$$

0 2

$$C_{31} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = 1$$

0 1

$$C_{32} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = -(-1) = 1$$

3 2

$$C_{33} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = -1$$

$$C = \begin{bmatrix} -2 & -1 & 2 \\ 1 & -2 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} -2 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$$

$$A' = \frac{1}{1} \begin{bmatrix} -2 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$$

$$0.2 \cdot A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix} \quad \begin{matrix} 1 & 2 \\ 0 & 1 \\ 2 & 1 \end{matrix}$$

2 0 0 3 0 0

$$\det(A) = 3 - 2 = 1$$

$$C_{11} = \begin{bmatrix} 1 & 0 \\ 1 & 3 \end{bmatrix} = 3$$

$$C_{12} = \begin{bmatrix} 0 & 0 \\ 2 & 3 \end{bmatrix} = -(0) = 0$$

$$C_{13} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = -2 \quad C_{21} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = -(5) = -5$$

$$C_{22} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = 1 \quad C_{23} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = -(1 \cdot 3) = -3$$

$$C_{31} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = 1 \quad C_{32} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = -(0) \cdot 0 \quad C_{33} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = 1$$

$$C = \begin{bmatrix} 3 & 0 & -2 \\ -5 & 1 & 3 \\ -1 & 0 & 1 \end{bmatrix} \quad \text{Adj}(A) = \begin{bmatrix} 3 & -5 & -1 \\ 0 & 1 & 0 \\ -2 & 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -5 & -1 \\ 0 & 1 & 0 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -5 & -1 \\ 0 & 1 & 0 \\ -2 & 3 & 1 \end{bmatrix}$$

Transformações Lineares

01. $f(x) = 2x + 1$

$$f(0) = 2 \cdot 0 + 1$$

$$f(0) = 1$$

$0 \neq 1$ (não é linear)

02. $f(x) = 5x$

$$f(ax) = 5ax = a(5x) = a f(x)$$

$$f(x+y) = 5(x+y) = 5x + 5y = f(x) + f(y)$$

$f(x) = 5x$ (é linear)

03. $f(x) = -x$

$$f(ax) = -ax = a(-x) = a f(x)$$

$$f(x+y) = -(x+y) = -x - y = f(x) + f(y)$$

$f(x) = -x$ (é linear)

04. $f(x) = x^2$

$$f(x+y) = (x+y)^2 = x^2 + 2xy + y^2 \neq f(x) + f(y) = x^2 + y^2$$

$f(x) = x^2$ (não é linear)

Representação matricial de transformações lineares

01. $x' = -3x + 2y$ e $y' = -2x + 2y$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

02. $x' = -5x + 3y$, $y' = -4x + 7y + 9z$ e $z' = 2y$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} -5 & 3 & 0 \\ -4 & 7 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

03. $x' = -y + z$, $y' = -x + 9z$ e $z' = 2y + 2x$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Transformação de escala

01. Amplie duas vezes o ponto $a = (2, 4, -6)$

$$a' = (2 \times 2, 4 \times 2, -6 \times 2)$$

$$a' = (4, 8, -12)$$

02. Reduza por um terço as coordenadas do ponto $a = (5, 3, 6)$.

$$a' = \left(\frac{1}{3} \times 5, \frac{1}{3} \times 3, \frac{1}{3} \times 6 \right)$$

$$a' = \left(\frac{5}{3}, 1, 2 \right)$$

Transformação de rotação

01. $a = (10, 12)$ rotacionando 30° .

$$x' = 10 \left(\frac{\sqrt{3}}{2} \right) - 12 \left(\frac{1}{2} \right)$$

$$x' = \frac{10\sqrt{3}}{2} - \frac{12}{2}$$

$$x' = 5\sqrt{3} - 6$$

$$y' = 12 \left(\frac{\sqrt{3}}{2} \right) + 10 \left(\frac{1}{2} \right)$$

$$y' = \frac{12\sqrt{3}}{2} + \frac{10}{2}$$

$$y' = 6\sqrt{3} + 5$$

$$a' = (5\sqrt{3} - 6, 6\sqrt{3} + 5)$$

02. $a = (3, -10)$ rotacionando 10°

$$x' = 3 \cos 10^\circ - (-10) \sin 10^\circ$$

$$x' = 3 \cos 10^\circ + 10 \sin 10^\circ$$

$$y' = (-10) \cos 10^\circ + 3 \sin 10^\circ$$

$$a' = (3 \cos 10^\circ + 10 \sin 10^\circ, -10 \cos 10^\circ + 3 \sin 10^\circ)$$

$$a' \approx (4,6904, -9,3272)$$

03. $a = (3, 7)$ rotacionando 45° ;

$$x' = 3 \left(\frac{\sqrt{2}}{2} \right) - 7 \left(\frac{\sqrt{2}}{2} \right)$$

$$x' = \frac{3\sqrt{2} - 7\sqrt{2}}{2} = \frac{-4\sqrt{2}}{2} = -2\sqrt{2}$$

$$y' = 7 \left(\frac{\sqrt{2}}{2} \right) + 3 \left(\frac{\sqrt{2}}{2} \right)$$

	30°	45°	60°
sen	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

$$y' = \frac{7\sqrt{2} + 3\sqrt{2}}{2} = \frac{10\sqrt{2}}{2} = 5\sqrt{2}$$

$$a' = (-2\sqrt{2}, 5\sqrt{2})$$

04. $a = (1, 1)$ rotacionando 60° ;

$$x' = 1\left(\frac{1}{2}\right) - 1\left(\frac{\sqrt{3}}{2}\right)$$

$$x' = \frac{1}{2} - \frac{\sqrt{3}}{2} = \frac{1-\sqrt{3}}{2}$$

$$y' = 1\left(\frac{1}{2}\right) + 1\left(\frac{\sqrt{3}}{2}\right)$$

$$y' = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1+\sqrt{3}}{2}$$

$$a' = \left(\frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$$